

Software Requirements Specification

Workflow Queue (WFQ) Administrator Application

Ivonne Arizpe

Hector Corona

Ruby Quinones

Janise Williams

CS3338 Course Group 9

Version 1.0
May 14, 2026

Contents

1	Introduction	4
1.1	Purpose	4
1.2	Scope	4
1.3	Intended Audience	4
1.4	Definitions and Abbreviations	4
2	Overall Description	5
2.1	Product Overview	5
2.2	Product Functions	5
2.3	User Classes	5
2.4	Operating Environment	5
2.5	Assumptions and Dependencies	6
3	Requirements Specification	6
3.1	Snapshot 2: Email Notification System	6
3.1.1	Task Assignment Notifications	6
3.1.2	Email Format Requirements	6
3.1.3	Notification Suppression and Error Handling	7
3.2	Snapshots 3 and 4: Workflow Export and Report Generation	7
3.2.1	PDF Export Requirements	7
3.2.2	CSV Export Requirements	7
3.2.3	Export Behavior and Security Requirements	8
4	Non-Functional Requirements	8
4.1	Performance	8
4.2	Security	8
4.3	Reliability	8
4.4	Usability	9
A	TestRail Test Case Reference	9
B	TestRail Test Run Summary	10

Revision History

Name	Date	Reason For Changes	Version
Ruby Quinones	5/14/2026	Initial Document	1.0
Hector Corona	5/14/2026	Section 1 Introduction	1.0
Ivonne Arizpe	5/14/2026	Section 2 Overall Description	1.0
Janise Williams	5/14/2026	Sections 3 and 4 Requirements	1.0

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Workflow Queue (WFQ) Administrator Application, as developed and tested by CS3338 Group 9. The requirements in this document are derived from the test cases written and executed by our team in TestRail across Snapshots 2, 3, and 4.

Each requirement in this document reflects an expected system behavior that our team identified, wrote a test case for, and validated during our test runs.

1.2 Scope

The WFQ Administrator Application is a web-based tool that allows authorized administrators to manage workflows, steps, tasks, and reports. This SRS covers the two major features tested by our team:

- **Email Notification System** (Snapshot 2): Automated email alerts sent to users and administrators when key workflow events occur.
- **Workflow Export and Report Generation** (Snapshots 3 and 4): The ability to export workflow data as a PDF or CSV file directly from the Workflow Details page.

1.3 Intended Audience

- **Group 9 Developers:** Use this document as the basis for implementing the features tested in TestRail.
- **QA Testers:** Use this document to verify that test cases align with defined requirements.
- **Course Instructor:** Use this document to evaluate Group 9's understanding of requirements derived from their own testing work.

1.4 Definitions and Abbreviations

- **WFQ** – Workflow Queue
- **SRS** – Software Requirements Specification
- **PDF** – Portable Document Format
- **CSV** – Comma-Separated Values
- **SMTP** – Simple Mail Transfer Protocol, used for sending emails
- **TestRail** – The test case management tool used by our team
- **Snapshot** – A project milestone in the CS3338 course timeline

- **Toast** – A brief, non-blocking UI notification message

2 Overall Description

2.1 Product Overview

The WFQ Administrator Application provides a web-based interface for administrators to create and manage workflows. In addition to core workflow management, our team designed and tested two extended features: an email notification system and a workflow export system. Both features were validated through TestRail test runs.

2.2 Product Functions

The features covered by this SRS are:

- Sending automated email notifications to users when workflow tasks are assigned, re-assigned, or completed.
- Sending automated email notifications to administrators when new workflow steps are created.
- Exporting a workflow's details as a formatted PDF file.
- Exporting a workflow's task list as a CSV file.
- Enforcing admin authorization for all export and notification-related actions.

2.3 User Classes

- **Administrators:** Have full access to workflow management, task oversight, and export features. Receive email notifications when tasks are completed and steps are created.
- **Assigned Users:** Receive email notifications when tasks are assigned or reassigned to them.
- **QA Testers (Group 9):** Responsible for executing test cases in TestRail and recording Pass/Fail results.

2.4 Operating Environment

- **Web Browser:** The application runs in modern browsers including Chrome, Edge, and Firefox.
- **Email:** Notifications are sent via an SMTP server configured in the application settings.
- **File Downloads:** PDF and CSV exports are delivered as browser file downloads.

- **PDF Viewer:** Exported PDFs must be readable in standard viewers such as Adobe Acrobat and browser-based PDF viewers.
- **Spreadsheet Tools:** Exported CSVs must be compatible with Microsoft Excel and Google Sheets.

2.5 Assumptions and Dependencies

- An SMTP server is available and configured in `appsettings.json`.
- Admin authorization is determined by the URL query string parameter `?is_admin=true`.
- The database contains at least one workflow, with steps and tasks, for export testing purposes.
- TestRail is connected to the project and test runs have been created for each snapshot.

3 Requirements Specification

3.1 Snapshot 2: Email Notification System

The following requirements are derived from the ten test cases written and executed by our team in the Snapshot 2 TestRail test run.

3.1.1 Task Assignment Notifications

- **REQ-2.1:** The system shall send an email notification to the assigned user when a task is assigned to them. (Derived from T-01)
- **REQ-2.2:** When a task is reassigned, the system shall send an email notification to the new assignee only. The previous assignee shall not receive a notification. (Derived from T-02)
- **REQ-2.3:** The system shall send an email notification to the workflow administrator when a task is marked as completed. (Derived from T-03)
- **REQ-2.4:** The system shall send an email notification to the workflow administrator when a new step is created within a workflow. (Derived from T-04)

3.1.2 Email Format Requirements

- **REQ-2.5:** Each notification email subject line shall follow the format: `[WFQ] Task Assigned: {TaskName}`. (Derived from T-07)
- **REQ-2.6:** Each notification email body shall include a valid URL link to the relevant workflow details page. (Derived from T-08)

3.1.3 Notification Suppression and Error Handling

- **REQ-2.7:** The system shall not send any email notification when a read-only action is performed, such as viewing a workflow or step. (Derived from T-09)
- **REQ-2.8:** If the SMTP server fails during a notification send, the system shall log the error and allow the original user action to complete successfully. No error message shall be shown to the user. (Derived from T-05)
- **REQ-2.9:** The notification system shall be toggleable via a configuration flag. When set to `Notifications:Enabled: false`, no emails shall be sent. (Derived from T-06)
- **REQ-2.10:** A completed task shall not be eligible for reassignment. The reassign button shall be disabled or hidden for completed tasks. (Derived from T-10)

3.2 Snapshots 3 and 4: Workflow Export and Report Generation

The following requirements are derived from the thirteen test cases written and executed across the Snapshot 3 and Snapshot 4 TestRail test runs. Snapshot 4 served as a final validation of the same feature set.

3.2.1 PDF Export Requirements

- **REQ-3.1:** The system shall allow an administrator to export a workflow as a PDF file from the Workflow Details page. (Derived from T-01)
- **REQ-3.2:** The exported PDF shall include the workflow name, a table of all steps, and a table of all tasks. (Derived from T-01)
- **REQ-3.3:** The PDF file shall be named using the format: `{WorkflowName}_Report_{YYYY-MM-DD}.p` (Derived from T-03)
- **REQ-3.4:** If a workflow has no steps, the PDF shall still download successfully and display the message: “No steps have been added to this workflow.” (Derived from T-05)
- **REQ-3.5:** The exported PDF shall be readable in standard PDF viewers including Adobe Acrobat and browser-based PDF viewers. (Derived from T-10)

3.2.2 CSV Export Requirements

- **REQ-3.6:** The system shall allow an administrator to export a workflow’s task list as a CSV file from the Workflow Details page. (Derived from T-02)
- **REQ-3.7:** The exported CSV shall contain the correct column headers and all relevant task data rows. (Derived from T-02)
- **REQ-3.8:** The CSV file shall be named using the format: `{WorkflowName}_Tasks_{YYYY-MM-DD}.cs` (Derived from T-04)

- **REQ-3.9:** If a workflow has no tasks, the CSV shall still download successfully and contain only the header row with no data rows. (Derived from T-06)
- **REQ-3.10:** The exported CSV shall open correctly in both Microsoft Excel and Google Sheets without formatting errors. (Derived from T-11)

3.2.3 Export Behavior and Security Requirements

- **REQ-3.11:** After an export is triggered, the administrator shall remain on the Workflow Details page. The file download shall trigger in the browser without navigating away from the page. (Derived from T-09)
- **REQ-3.12:** Export actions shall not create, update, or delete any records in the database. All export operations are strictly read-only. (Derived from T-07)
- **REQ-3.13:** Export functionality shall require admin authorization. Any attempt to access an export URL without `?is_admin=true` shall return an Access Denied response. (Derived from T-08)
- **REQ-3.14:** While an export is being generated, the export button shall display a loading spinner and become disabled to prevent duplicate requests. (Derived from T-12)
- **REQ-3.15:** If the export fails due to a server error, the system shall display a red error toast notification. The page shall remain functional after the error. (Derived from T-13)

4 Non-Functional Requirements

4.1 Performance

- PDF and CSV generation shall complete within 3 seconds for workflows containing up to 100 steps and 500 tasks under normal server load.
- Email notifications shall be sent asynchronously and shall not delay the completion of the user's primary action.

4.2 Security

- All export and notification actions shall require admin authorization via the `?is_admin=true` URL query string parameter.
- SMTP credentials shall be stored in environment variables and shall never be hardcoded or committed to source control.

4.3 Reliability

- The system shall continue normal operation if an email notification fails to send.

- The system shall continue normal operation if an export encounters a server error, and shall display an appropriate error message to the user.

4.4 Usability

- Error messages shall be user-friendly and shall not expose raw system or database details to the user.
- The export button shall provide visual feedback in the form of a loading spinner during file generation.

A TestRail Test Case Reference

Requirement	TestRail ID	Test Case Title
REQ-2.1	Snapshot 2, T-01	Task assigned — notification sent
REQ-2.2	Snapshot 2, T-02	Task reassigned — new assignee notified
REQ-2.3	Snapshot 2, T-03	Task completed — admin notified
REQ-2.4	Snapshot 2, T-04	Step created — admin notified
REQ-2.5	Snapshot 2, T-07	Email subject format correct
REQ-2.6	Snapshot 2, T-08	Email body contains workflow link
REQ-2.7	Snapshot 2, T-09	Read-only action — no notification sent
REQ-2.8	Snapshot 2, T-05	SMTP failure — operation still succeeds
REQ-2.9	Snapshot 2, T-06	Notifications disabled flag — no email sent
REQ-2.10	Snapshot 2, T-10	Completed task cannot be reassigned
REQ-3.1	Snapshot 3, T-01	Export PDF — workflow with steps and tasks
REQ-3.2	Snapshot 3, T-01	Export PDF — content includes steps and tasks
REQ-3.3	Snapshot 3, T-03	PDF file name format correct
REQ-3.4	Snapshot 3, T-05	Export PDF — workflow with no steps
REQ-3.5	Snapshot 3, T-10	PDF readable in standard PDF viewer
REQ-3.6	Snapshot 3, T-02	Export CSV — workflow with tasks
REQ-3.7	Snapshot 3, T-02	Export CSV — correct columns and data
REQ-3.8	Snapshot 3, T-04	CSV file name format correct
REQ-3.9	Snapshot 3, T-06	Export CSV — workflow with no tasks
REQ-3.10	Snapshot 3, T-11	CSV compatible with Excel and Google Sheets
REQ-3.11	Snapshot 3, T-09	Admin stays on page after export
REQ-3.12	Snapshot 3, T-07	Export does not modify database
REQ-3.13	Snapshot 3, T-08	Export requires admin authorization
REQ-3.14	Snapshot 3, T-12	Export button shows loading state
REQ-3.15	Snapshot 3, T-13	Export error shows toast notification

B TestRail Test Run Summary

Snapshot	Feature	Total	Passed	Failed
Snapshot 2	Email Notification System	10	6	4
Snapshot 3	Workflow Export (PDF and CSV)	13	9	2
Snapshot 4	Final Validation (PDF and CSV)	13	10	3